



# Follow-up Observations and an Improved Orbital Period for Exoplanet Candidate TOI-3505.01

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## Abstract

TOI-3505.01 is a planet candidate found by NASA's TESS mission. It circles a 6220 K star that is 1.34 times the radius of the Sun (TIC 390988385,  $V = 11.2$ ) every 2.9151 days. On July 22, 2022 we watched the star with the 0.8 m telescope at George Mason University during the hours the transit was supposed to happen. We took 283 images in R band and kept 263 good measurements, with a scatter of 2.50 ppt. The star never dimmed ( $-0.66 \pm 0.39$  ppt). To see whether that meant anything, we added fake transits to our data. A fake 2.91 ppt transit came back at  $5.7\sigma$ , so we would have caught a real one.

We saw nothing because the prediction was off. With the period we have now, the closest transit was 17.4 hours before we started. So we went to the TESS archive and timed 25 transits in Sectors 14, 41, 54, and 81, covering 5.05 years. Fitting a line through those times gives a period of  $2.9151503 \pm 0.0000057$  days, twice as precise as the TESS catalog value and matching it to within  $0.4\sigma$ . It also tightens the predicted transit time for 2026 from 9 minutes to 3.

## Materials and Methods

### Taking and measuring the images

- Mason 0.8 m telescope, July 22, 2022 UT. R filter, 50 second exposures,  $0.362''$  per pixel. We got 283 images and 281 of them lined up into one 5.07 hour run.
- We subtracted darks, divided by flats, and shifted every image so the stars matched, using 25 reference stars, then redid each step in Python to check it. Astrometry.net matched our field to sky coordinates and landed 0.47 px from our measured position.
- We measured brightness in AstrolImageJ with a **25 px (9.1'') aperture**, a **70–139 px sky ring**, and comparison stars C2–C11. We picked 25 px by trying every radius from 20 to 50 px and keeping the one with the least scatter. Our own Python photometry agrees with AstrolImageJ at  $r = 0.990$ .
- We cut 18 measurements for bad image quality and wrote down why for each, leaving **263**. No detrending passed the test we set up ahead of time, so the light curve is left as measured.

### Working with the TESS archive

- We pulled QLP light curves for Sectors 14, 41, 54, and 81, plus the SPOC light curves and the official Data Validation reports.
- We ran a box least-squares search on each sector by itself, then fit each transit one at a time to get 25 mid-transit times and fit a straight line through them.
- We used the TESS pixel data to build difference images, find where on the sky the dimming happens, and check how much light nearby stars add. We locked in our cuts before looking at the final numbers.

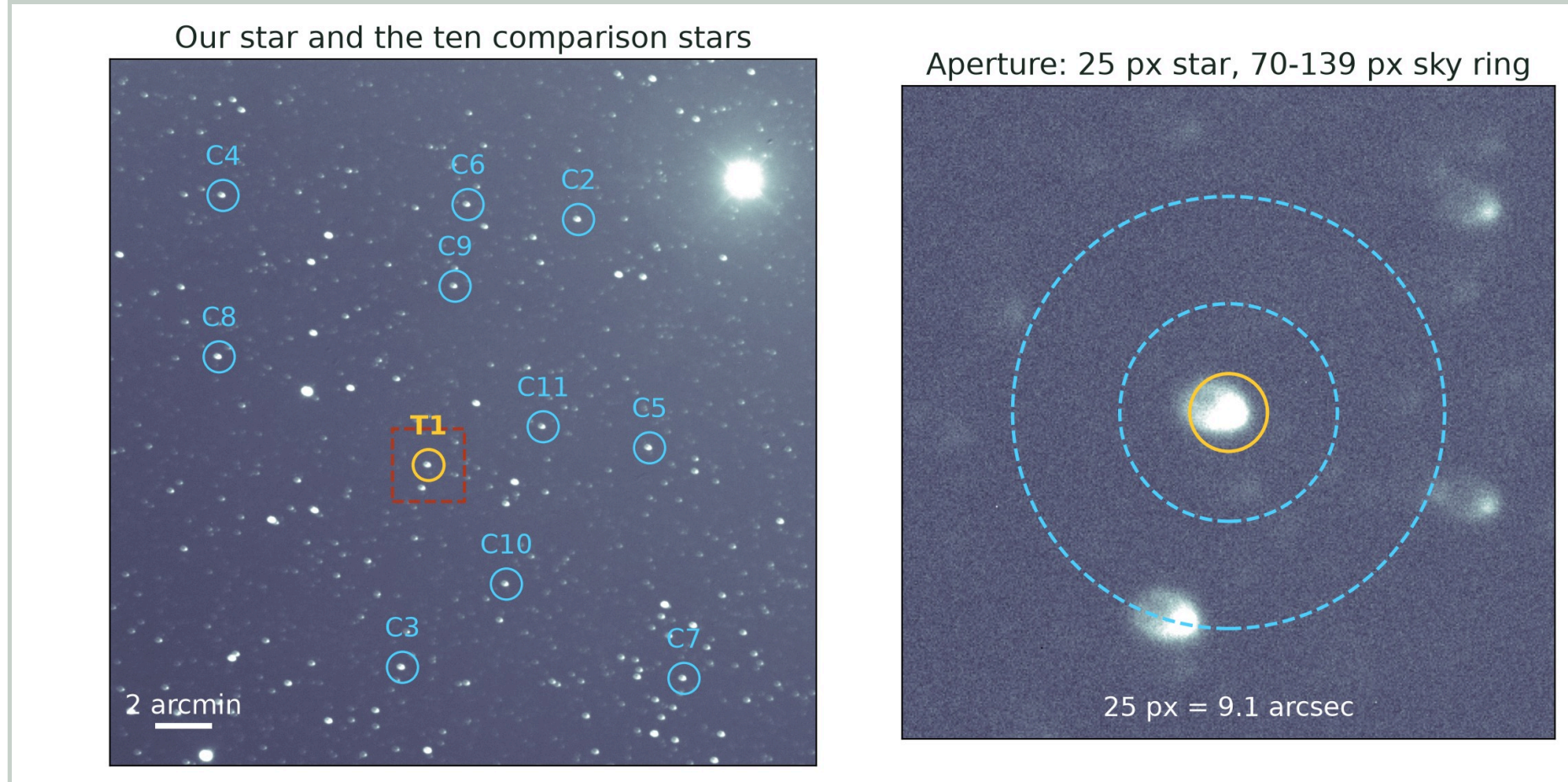
## System Parameters

| The star, TIC 390988385                                                                           |                                              |
|---------------------------------------------------------------------------------------------------|----------------------------------------------|
| Temperature, radius, mass                                                                         | 6220 K, $1.335 R_{\odot}$ , $1.25 M_{\odot}$ |
| Distance, TESS and V magnitude                                                                    | 373 pc, 10.94, 11.24                         |
| Nearby star, extra light                                                                          | $0.517''$ ( $\Delta I = 1.7$ ), 0.55         |
| What we measured                                                                                  |                                              |
| Orbital period                                                                                    | $2.9151503 \pm 0.0000057$ d                  |
| Reference time, BJD <sub>TDB</sub>                                                                | $2459793.5331 \pm 0.0013$                    |
| Depth and length, 4 sectors                                                                       | 3.05 ppt, 2.06 h                             |
| Depth during our window                                                                           | $-0.66 \pm 0.39$ ppt                         |
| What that gives for the planet                                                                    |                                              |
| Size next to the star, $R_p/R_{\star}$                                                            | $0.0618 \pm 0.0016$                          |
| Planet radius                                                                                     | $9.0 R_{\oplus} = 0.80 R_J$                  |
| How centered the crossing is, tilt                                                                | $b = 0.85$ , $82.9^\circ$                    |
| $a/R_{\star}$ , orbit size, temperature                                                           | 6.93, $0.043$ au, 1670 K                     |
| Based on the TIC v8 star radius and the SPOC depth ratio, with no correction for the nearby star. |                                              |

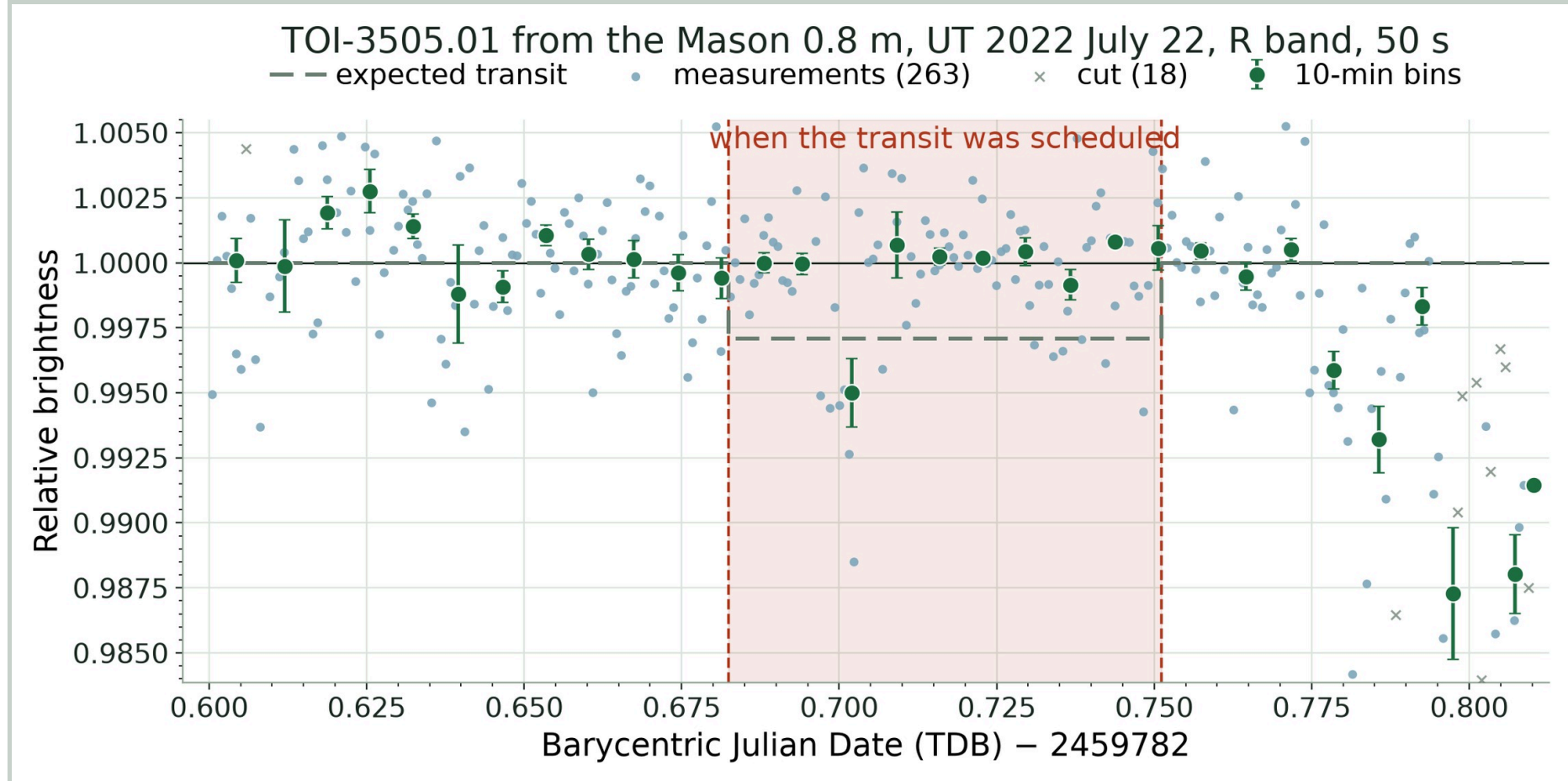
## Future Work

- Watch a transit using our new period. A 3 minute window makes that worth doing on a 0.8 m telescope.
- Fit a real limb-darkened model with MCMC and a density prior to pin down  $b$  and the planet's size.
- Observe in more than one filter, since a depth that changes with color would mean a blended eclipsing binary. Measure how bright the nearby star is in the TESS band, and take radial velocities to weigh the planet.

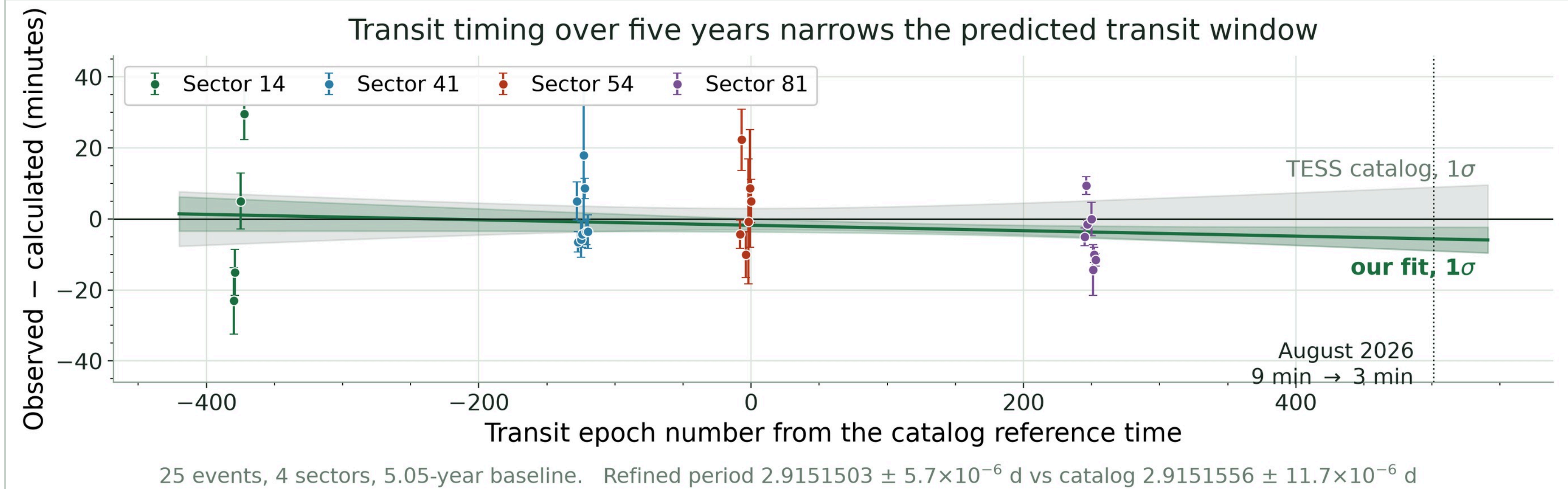
## Results



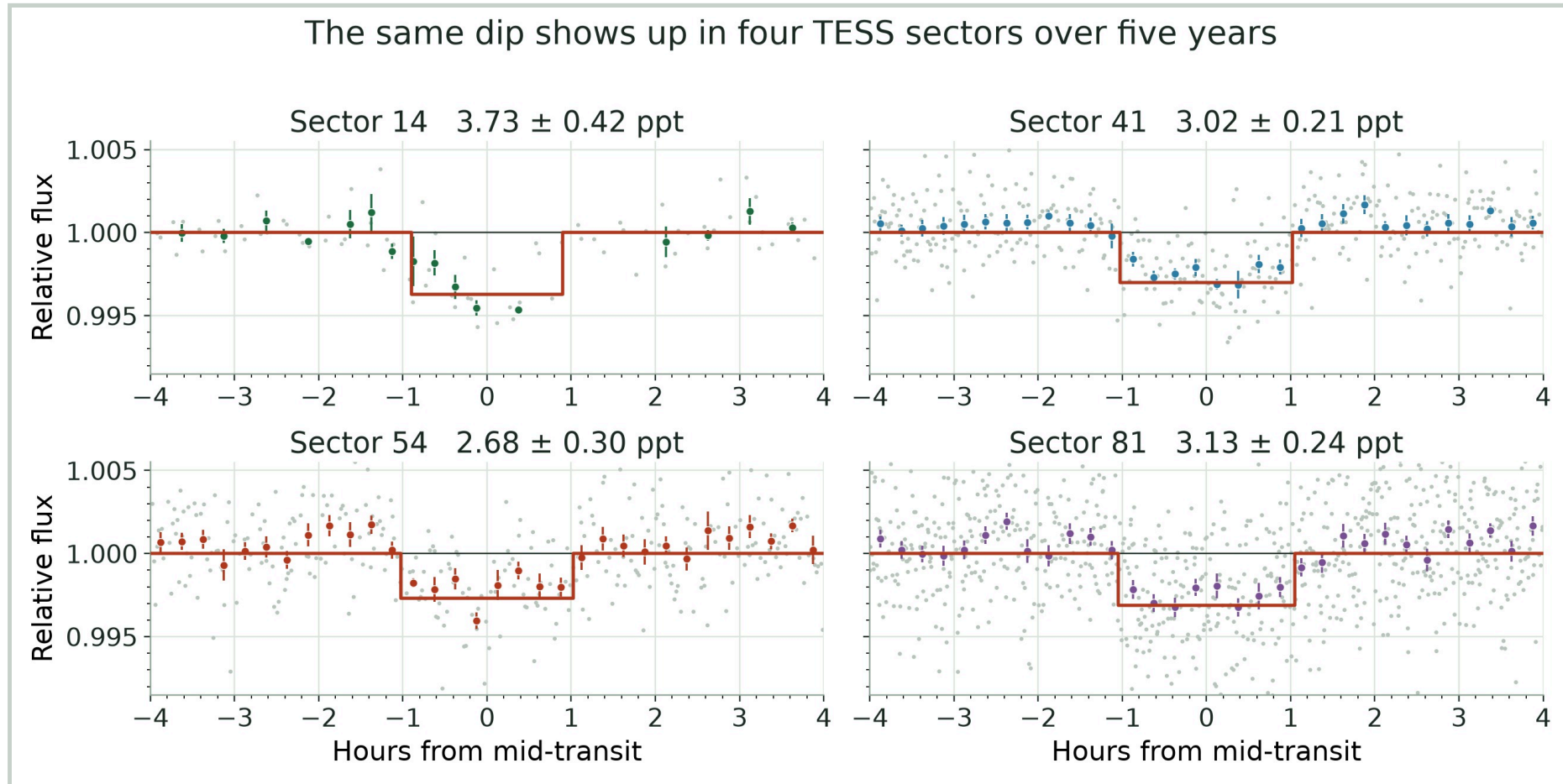
**Fig. 1.** The field around TOI-3505 after we matched it to sky coordinates. T1 is our star and C2–C11 are the comparison stars. The right panel zooms in on the 25 px aperture and the 70–139 px sky ring we used.



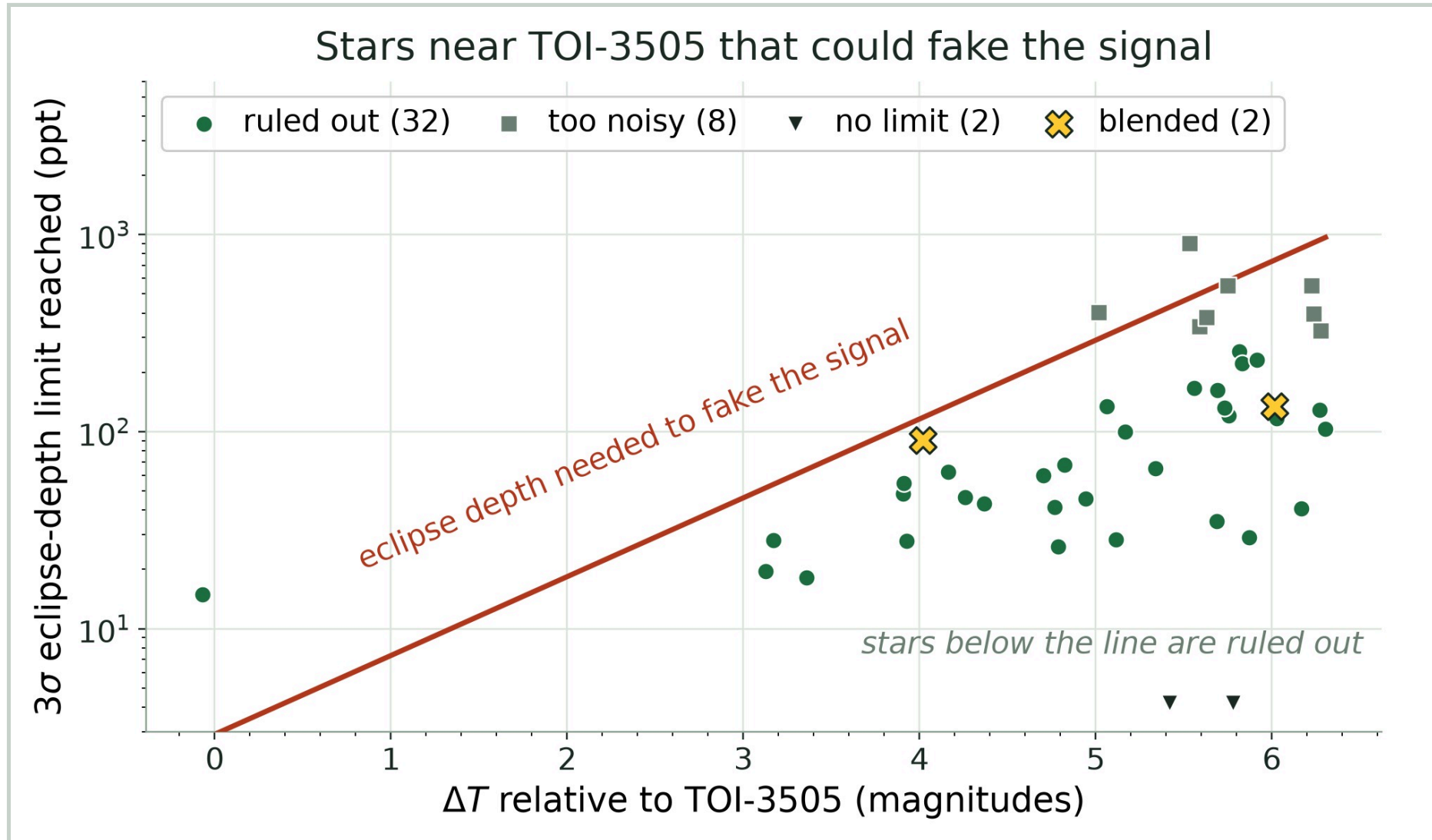
**Fig. 2.** Our final light curve: 263 measurements, 2.50 ppt scatter, no detrending. The dashed line shows the dip we would have seen if the transit had happened on schedule. Nothing is there. The points get noisier at the end of the night as the star dropped closer to the horizon.



**Fig. 3. Our main result.** Each point is one transit we timed, compared with the time the TESS catalog predicts. Grey is the catalog's  $1\sigma$  range and green is ours. Because our transits stretch across 5.05 years, our period is twice as precise, and the predicted transit time for 2026 tightens from 9 minutes to 3. We widened the error bars by 2.30 first, so this is the careful version rather than the best-case one.



**Fig. 4.** TESS data folded on our new period. The dip shows up on its own in all four sectors across five years, and the four depths agree ( $\chi^2 = 4.3$  for 3 degrees of freedom).



**Fig. 5.** The 44 catalog stars within  $60''$  of our target. The red line is how deep an eclipse each one would need to fake our signal. A star is ruled out if its own light curve is good enough to say that eclipse did not happen. 32 are ruled out. The rest are too faint to measure well or sit too close to our star.

### Our numbers next to the published ones

| Quantity               | Ours — S14, 41, 54, 81    | TESS catalog — S14, 41, 54 | SPOC — S54, 81            |
|------------------------|---------------------------|----------------------------|---------------------------|
| Orbital period (days)  | $2.9151503 \pm 0.0000057$ | $2.9151556 \pm 0.0000117$  | $2.9151456 \pm 0.0000070$ |
| Transit depth (ppt)    | $3.05 \pm 0.13$           | $2.91 \pm 0.20$            | $3.29 \pm 0.12$           |
| Transit length (hours) | $2.06 \pm 0.05$           | $2.00 \pm 0.21$            | $2.71 \pm 0.10$           |

## Discussion

### Why we saw no transit

We picked this night using a 2022 prediction. With the period we have now, the nearest transit was 17.4 hours before we started, so there was nothing to catch. Our run also lands in a gap in TESS Sector 54, so we cannot line the two up. We added fake 2.91 ppt transits to our data and pulled them back out at  $5.7\sigma$ , so a real transit would have shown up. Our flat light curve is a real limit and not just noise.

### What the TESS archive gave us

One night could not fix the prediction, so we used the archive instead. Timing 25 transits across four sectors gives  $P = 2.9151503 \pm 0.0000057$  d. That is 2.05 times more precise than the catalog and agrees with it to  $0.4\sigma$ . The timing points scatter more than their error bars allow ( $\chi^2/\text{dof} = 5.3$ , RMS 11.8 min). That is what we expect from Sector 14, where each image covers 30 minutes, so we widened the error bars instead of claiming the tighter number. Leaving Sector 14 out shifts the period by much less than  $1\sigma$ .

### Why the planet's size is uncertain

In the official SPOC fit the planet barely clips the edge of the star ( $b = 0.92$ ). For that fit to work, the star would have to be 3.6 times less dense than a  $1.34 R_{\odot}$ ,  $1.25 M_{\odot}$  star can be. When a transit is this close to the edge, the depth, the length,  $b$ , and  $a/R_{\star}$  all trade off, so lots of combinations fit the same data. If we keep the star's density and match our own 2.06 h transit length, we get  $b = 0.85$  and  $a/R_{\star} = 6.93$ . Either way the planet only grazes the star, so  $9.0 R_{\oplus}$  is much less certain than its error bar makes it look.

### What we cannot claim

Another star sits  $0.517''$  away and adds about 21 percent of the primary's light. Our telescope cannot separate the two and neither can TESS, so if the transit belongs to the brighter star, the real depth and radius are larger. Our SPOC comparisons use the same TESS data we used, so they test our method rather than give us a second detection. We improved the orbit and ruled out several false alarms, but we did not confirm a planet.

## Checks for False Alarms

|                                   |                      |
|-----------------------------------|----------------------|
| Same depth in all four sectors    | $\chi^2 = 4.3 / 3$   |
| Odd and even transits match       | $\leq 1.7\sigma$     |
| No second dip half an orbit later | $S/N \leq 1.7$       |
| Dimming centered on our star      | $1.6\text{--}5.1''$  |
| Compared with one TESS pixel      | inside $21''$        |
| Nearby stars ruled out            | 32 of 44             |
| SPOC odds of a false alarm        | $1 \times 10^{-143}$ |
| SPOC ghost-image check            | 6.9                  |
| Star $0.517''$ away               | cannot separate      |

## Conclusions

- A better period.** Timing 25 transits over 5.05 years gives  $P = 2.9151503 \pm 0.0000057$  d, twice as precise as the catalog value.
- Better predictions.** The transit time for 2026 is now good to 3 minutes instead of 9, so the next person to observe needs much less time on either side of it.
- Our night still counts.** The 2.50 ppt light curve rules out a transit during the window we watched, and our fake-transit tests show we would have seen one.
- Still a good candidate.** Nothing we checked points to a false alarm. The signal looks like a roughly Jupiter-sized planet grazing the edge of its star.
- What holds us back.** The grazing path and the nearby star limit the size measurement now, not our photometry.

## Acknowledgements

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## Major Citations

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